Mathematics Method for Physics—Lecture 8

Laplace Transform: Definition and Properties

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Conventions: $p = \sigma + \mathrm{i}\omega$; transform pairs are written function first, $\{f(t), F(p)\}$; $\mathrm{e}, \mathrm{i}, \mathrm{d}$ upright; formulas are numbered (8.1)–(8.24).

8 Laplace Transform

8.1 Definition

For $f(t)$ defined on $[0, +\infty)$ put

$$F(p) = \int_0^{+\infty} f(t)\, \mathrm{e}^{-pt}\, \mathrm{d}t, \tag{8.1}$$

where

$$p = \sigma + \mathrm{i}\omega \tag{8.2}$$

is a **complex variable**. Then $\{f(t), F(p)\}$ is a **Laplace transform pair**, written

$$\mathcal{L}\left[f(t)\right] = F(p), \qquad \text{or} \qquad \mathcal{L}^{-1}\left[F(p)\right] = f(t). \tag{8.3}$$

Remarks

1. $F(p)$ is complex-valued. The imaginary part ω is exactly the Fourier frequency, the real part σ acts as a **damping factor**: multiplying f by $\mathrm{e}^{-\sigma t}$ before transforming is why the Laplace transform reaches functions of moderate growth that the Fourier transform does not.
2. The lower limit is written 0; for a function with an impulse at $t = 0$ the left limit 0^- is meant — with 0^- one gets $\mathcal{L}[\delta] = 1$, with 0^+ one gets 0.
3. Only $t \geq 0$ enters (8.1), so the values of $f(t)$ for $t < 0$ do not affect $F(p)$: what is transformed is the causal part $f(t)u(t)$.

8.2 Exponential order and the existence theorem

8.2.1 Functions of exponential order

Definition 8.1 (Exponential order)**.** If there exist constants $M > 0$ and $\sigma_c \geq 0$ such that

$$|f(t)| \leq M\,\mathrm{e}^{\sigma_c t}, \qquad t \geq 0, \tag{8.4}$$

then $f(t)$ is said to be of **exponential order**, and σ_c is called its **growth exponent**.

- $u(t)$, t^n, e^{at}, $\sin\beta t$, $\cos\beta t$ are all of exponential order;
- $t^{-1/2}$ is unbounded as $t \to 0^+$ and is **not** of exponential order, yet $\int_0^{+\infty} t^{-1/2}\mathrm{e}^{-pt}\,\mathrm{d}t$ still converges (integrable singularity);
- e^{t^2} grows faster than any exponential function and is not of exponential order — its Laplace transform does not exist. Exponential order is therefore **sufficient, not necessary**.

8.2.2 Theorem 8.1 (existence theorem)

Theorem 8.1. *Let $f(t)$ on $[0, +\infty)$ satisfy*

1. *the **Dirichlet conditions** (piecewise continuous on every finite interval, finitely many discontinuities of the first kind, finitely many extrema), and*
2. *$|f(t)| \leq M\,\mathrm{e}^{\sigma_c t}$ (exponential order).*

Then

$$F(p) = \int_0^{+\infty} f(t)\,\mathrm{e}^{-pt}\,\mathrm{d}t \tag{8.5}$$

*exists and is **analytic** on the half-plane $\operatorname{Re} p > \sigma_c$.*

8.2.3 Proof

1. Absolute convergence. For $\sigma = \operatorname{Re} p > \sigma_c$, using $\left|\mathrm{e}^{-pt}\right| = \mathrm{e}^{-\sigma t}$,

$$\int_0^{+\infty} \left|f(t)\,\mathrm{e}^{-pt}\right| \mathrm{d}t \leq \int_0^{+\infty} M\,\mathrm{e}^{(\sigma_c - \sigma)t}\,\mathrm{d}t = \frac{M}{\sigma - \sigma_c} < +\infty. \tag{8.6}$$

2. Uniform convergence and differentiability. On $\operatorname{Re} p \geq \sigma_0 > \sigma_c$ the integral (8.5) converges uniformly, and so does the differentiated integral $\int_0^{+\infty} f(t)\,(-t)\,\mathrm{e}^{-pt}\,\mathrm{d}t$; differentiation under the integral sign is therefore legitimate:

$$F'(p) = \int_0^{+\infty} f(t)\left(\mathrm{e}^{-pt}\right)'_p \mathrm{d}t = -\int_0^{+\infty} t\,f(t)\,\mathrm{e}^{-pt}\,\mathrm{d}t. \tag{8.7}$$

So F is differentiable — hence analytic — on $\operatorname{Re} p > \sigma_c$. ∎

8.2.4 Remarks

1. $F(p)$ is analytic for $\operatorname{Re} p > \sigma_c$, and

$$\lim_{\operatorname{Re} p \to +\infty} F(p) = 0. \tag{8.8}$$

2. By **analytic continuation** $F(p)$ is analytic on the whole p-plane except at its singularities; the positions of the poles / branch points determine the structure of $f(t)$.

3. **Region of convergence.** The theorem gives $\operatorname{Re} p > \sigma_c$, i.e. the right half-plane $\sigma > \sigma_c$ (σ horizontal, $\mathrm{i}\omega$ vertical in the p-plane). It depends on f: $\mathcal{L}[u]$ converges for $\operatorname{Re} p > 0$, $\mathcal{L}[\mathrm{e}^{at}]$ for $\operatorname{Re} p > \operatorname{Re} a$; for a rational image the ROC is fixed by the rightmost singularity.

8.3 Common transform pairs

Every entry below is obtained by **direct integration** of (8.1).

8.3.1 $u(t) \leftrightarrow 1/p$

With $u(t) = 1$ for $t \geq 0$ and 0 for $t < 0$,

$$\mathcal{L}\left[u(t)\right] = \int_0^{+\infty} \mathrm{e}^{-pt}\,\mathrm{d}t = \left.\frac{\mathrm{e}^{-pt}}{-p}\right|_0^{+\infty} = \frac{1}{p}, \qquad \mathrm{Re}\,p > 0. \tag{8.9}$$

8.3.2 Powers t, t^2, ..., t^n

By parts ($u = t$, $\mathrm{d}v = \mathrm{e}^{-pt}\mathrm{d}t$):

$$\begin{aligned}\mathcal{L}\left[t\right] &= \int_0^{+\infty} t\,\mathrm{e}^{-pt}\,\mathrm{d}t = \left[t\left(-\frac{\mathrm{e}^{-pt}}{p}\right)\right]_0^{+\infty} + \int_0^{+\infty} \frac{\mathrm{e}^{-pt}}{p}\,\mathrm{d}t \\ &= \frac{1}{p^2},\end{aligned} \tag{8.10}$$

$$\mathcal{L}\left[t^2\right] = \left[t^2\left(-\frac{\mathrm{e}^{-pt}}{p}\right)\right]_0^{+\infty} + \frac{2}{p}\int_0^{+\infty} t\,\mathrm{e}^{-pt}\,\mathrm{d}t = \frac{2!}{p^3}, \tag{8.11}$$

and in general (by induction)

$$\mathcal{L}\left[t^n\right] = \frac{n!}{p^{n+1}}, \qquad \mathrm{Re}\,p > 0. \tag{8.12}$$

8.3.3 Exponential $\mathrm{e}^{at} \leftrightarrow 1/(p-a)$

$$\mathcal{L}\left[\mathrm{e}^{at}\right] = \int_0^{+\infty} \mathrm{e}^{(a-p)t}\,\mathrm{d}t = \left.\frac{\mathrm{e}^{(a-p)t}}{a-p}\right|_0^{+\infty} = \frac{1}{p-a}, \tag{8.13}$$

which converges for $\mathrm{Re}(a-p) < 0$, i.e. $\mathrm{Re}\,p > \mathrm{Re}\,a$.

8.3.4 Sine and cosine (via Euler's formula)

From $\sin \beta t = \dfrac{\mathrm{e}^{\mathrm{i}\beta t} - \mathrm{e}^{-\mathrm{i}\beta t}}{2\mathrm{i}}$, and using linearity (Property 1),

$$\mathcal{L}\left[\sin \beta t\right] = \frac{1}{2\mathrm{i}} \left(\frac{1}{p - \mathrm{i}\beta} - \frac{1}{p + \mathrm{i}\beta} \right) = \frac{1}{2\mathrm{i}} \cdot \frac{2\mathrm{i}\beta}{p^2 + \beta^2} = \frac{\beta}{p^2 + \beta^2}. \tag{8.14}$$

Similarly, from $\cos \beta t = \dfrac{\mathrm{e}^{\mathrm{i}\beta t} + \mathrm{e}^{-\mathrm{i}\beta t}}{2}$,

$$\mathcal{L}\left[\cos \beta t\right] = \frac{p}{p^2 + \beta^2}. \tag{8.15}$$

8.3.5 Hyperbolic functions

From $\operatorname{sh} \beta t = \dfrac{\mathrm{e}^{\beta t} - \mathrm{e}^{-\beta t}}{2}$ and $\operatorname{ch} \beta t = \dfrac{\mathrm{e}^{\beta t} + \mathrm{e}^{-\beta t}}{2}$,

$$\mathcal{L}\left[\operatorname{sh} \beta t\right] = \frac{\beta}{p^2 - \beta^2}, \qquad \mathcal{L}\left[\operatorname{ch} \beta t\right] = \frac{p}{p^2 - \beta^2}. \tag{8.16}$$

8.3.6 Table 8.1 — pairs used in practice

No.	$f(t)$, $t \geq 0$	$F(p) = \mathcal{L}[f(t)]$	ROC
1	$u(t)$	$\frac{1}{p}$	$\operatorname{Re} p > 0$
2	t	$\frac{1}{p^2}$	$\operatorname{Re} p > 0$
3	t^n	$\frac{n!}{p^{n+1}}$	$\operatorname{Re} p > 0$
4	e^{at}	$\frac{1}{p-a}$	$\operatorname{Re} p > \operatorname{Re} a$
5	$\sin \beta t$	$\frac{\beta}{p^2+\beta^2}$	$\operatorname{Re} p > 0$
6	$\cos \beta t$	$\frac{p}{p^2+\beta^2}$	$\operatorname{Re} p > 0$
7	$\operatorname{sh} \beta t$	$\frac{\beta}{p^2-\beta^2}$	$\operatorname{Re} p > \lvert\beta\rvert$
8	$\operatorname{ch} \beta t$	$\frac{p}{p^2-\beta^2}$	$\operatorname{Re} p > \lvert\beta\rvert$

Entry 4 and entries 5–6 are derived with Euler's formula; the rest follow by direct integration.

8.4 Properties 1–4

Throughout, $\{f(t), F(p)\}$, $\{f_1(t), F_1(p)\}$, $\{f_2(t), F_2(p)\}$ are transform pairs.

8.4.1 Property 1 — linearity

For constants a, b,

$$\mathcal{L}\left[af_1 + bf_2\right] = aF_1 + bF_2 \tag{8.17}$$

and the inverse is linear as well:

$$\mathcal{L}^{-1}\left[aF_1 + bF_2\right] = af_1 + bf_2. \tag{8.18}$$

Example 8.1 (Partial fractions)**.** Find $\mathcal{L}^{-1}\left[\frac{p}{(p-1)(p^2+4)}\right]$.

$$\frac{p}{(p-1)(p^2+4)} = \frac{A}{p-1} + \frac{Bp+C}{p^2+4} \quad \Longrightarrow \quad A = \frac{1}{5}, \qquad Bp + C = -\frac{1}{5}p + \frac{4}{5},$$

so, with Table 8.1 (entries 4, 5, 6),

$$\mathcal{L}^{-1}\left[\frac{p}{(p-1)(p^2+4)}\right] = \frac{1}{5}\mathrm{e}^{t} - \frac{1}{5}\cos 2t + \frac{2}{5}\sin 2t. \tag{8.19}$$

8.4.2 Property 2 — similarity

For $a > 0$,

$$\mathcal{L}\left[f(at)\right] = \frac{1}{a}F\left(\frac{p}{a}\right), \tag{8.20}$$

equivalently (replace a by $1/a$),

$$\mathcal{L}\left[\frac{1}{a}f\left(\frac{t}{a}\right)\right] = F(ap). \tag{8.21}$$

Proof (sketch). With $\tau = at$, $\mathrm{d}t = \mathrm{d}\tau/a$,

$$\mathcal{L}\left[f(at)\right] = \frac{1}{a}\int_0^{+\infty} f(\tau)\,\mathrm{e}^{-(p/a)\tau}\,\mathrm{d}\tau = \frac{1}{a}F\left(\frac{p}{a}\right).$$

8.4.3 Property 3 — delay (time shift of $f(t)u(t)$)

$$\mathcal{L}\left[f(t-t_0)\,u(t-t_0)\right] = \mathrm{e}^{-pt_0}F(p), \tag{8.22}$$

with inverse

$$\mathcal{L}^{-1}\left[\mathrm{e}^{-pt_0}F(p)\right] = f(t-t_0)\,u(t-t_0) \qquad (t_0>0). \tag{8.23}$$

Note. The shift must act on the *whole* function, i.e. $f(t-t_0)$ must carry $u(t-t_0)$; otherwise f is non-zero for $t<t_0$ and the transform is wrong.

Proof (sketch). With $\tau = t-t_0$,

$$\mathcal{L}\left[f(t-t_0)u(t-t_0)\right] = \int_{t_0}^{+\infty} f(t-t_0)\,\mathrm{e}^{-pt}\,\mathrm{d}t = \int_0^{+\infty} f(\tau)\,\mathrm{e}^{-p(\tau+t_0)}\,\mathrm{d}\tau = \mathrm{e}^{-pt_0}F(p).$$

8.4.4 Property 4 — frequency shift

$$\mathcal{L}\left[\mathrm{e}^{at}f(t)\right] = F(p-a). \tag{8.24}$$

Proof. Substitute $p \mapsto p-a$ in the definition:

$$F(p-a) = \int_0^{+\infty} f(t)\,\mathrm{e}^{-(p-a)t}\,\mathrm{d}t = \int_0^{+\infty}\left[\mathrm{e}^{at}f(t)\right]\mathrm{e}^{-pt}\,\mathrm{d}t = \mathcal{L}\left[\mathrm{e}^{at}f(t)\right].$$

Use. Together with Property 1 and Table 8.1 this is the "table + partial fractions + shift" inversion recipe, e.g.

$$\mathcal{L}^{-1}\left[\frac{1}{(p-a)^2}\right] = t\,\mathrm{e}^{at}, \qquad \mathcal{L}^{-1}\left[\frac{p-a}{(p-a)^2+\beta^2}\right] = \mathrm{e}^{at}\cos\beta t.$$

8.4.5 Table 8.2 — the four properties at a glance

Property	Content	No.
1 Linearity	$\mathcal{L}[af_1+bf_2] = a\mathcal{L}[f_1]+b\mathcal{L}[f_2]$; same for $\mathcal{L}^{-1}$	(8.17)(8.18)
2 Similarity	$\mathcal{L}[f(at)] = \frac{1}{a}F(\frac{p}{a})$, $a>0$	(8.20)
3 Delay	$\mathcal{L}[f(t-t_0)u(t-t_0)] = \mathrm{e}^{-pt_0}F(p)$	(8.22)
4 Frequency shift	$\mathcal{L}[\mathrm{e}^{at}f(t)] = F(p-a)$	(8.24)

8.5 Homework

- **A** 9, 10; **B** 1, 2, 5, 6, 8.